Innovation Centre for Education

YENEPOYA

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Project – LowLevel Design

on

Performance Evaluation of Natural Language Interface for E-commerce Infrastructure

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Table of Contents

Introduction	
2. Project Goal	
3. Model Development	
4. Process Flow Diagram	
5. Working	
6. Data Design	
7. Challenges	
8. References	

INTRODUCTION

The Natural Language Interface (NLI) for E-commerce allows users to interact with the platform using human language instead of traditional search filters. This project focuses on evaluating the performance of such an interface in terms of accuracy, response time, and user satisfaction.
The system converts user queries into structured database queries and retrieves relevant product information efficiently

PROJECT GOAL

The primary goals of the project are:

- To evaluate the accuracy of natural language query processing.
- To measure responce time and system efficiency.
- To analyze user intent recognition performance.
- To improve search relevance in e-commerce systems.

MODEL DEVELOPMENT

Components:

- **Data Preprocessing**
  - Tokenization of user input.
  - Stop-word removal.
- **NLP Processing Module**
  - Intent detection
  - Entity recognition(product,price,category)
- **Query Conversion Module**
  - Convert natural language → SQL/Structured query.

PROCESS FLOW DIAGRAM

1. User uploads a natural language query.
2. System process the query and converts it into a structured(SQL)query.
3. Database is required to fetch relevant results.
4. System measures accuracy and response time.
5. Final results are shown to the user.

User Input → Processing Module → NLP Engine → Query Conversion → Database Query Executuon → Result Retrieval → Performance Evaluation → Output to User

WORKING

1. User enters a natural language query.
2. System preprocesses the input text.
3. NLP model identifies intent and entities.
4. Query is converted into structured format(SQL).
5. Database fetches relevant products.
6. Results are ranked and displayed.
7. Performance metrics are recorded for evaluation.

DATA DESIGN

Dataset Fields:

- Product_ID
- Product_Name
- Category
- Price
- Brand
- Rating
- Availability
- Description

User Query Dataset:

- Query_ID
- User_Query
- Expected_Output
- Actual_Output
- Accuracy_Score
- Response_Time

CHALLENGES

1.Understanding ambiguous queries
2.Handling spelling mistakes
3.Complex query interpretation
4.Large dataset processing
5.Real-time response optimization
6.Multi-language support

REFERENCES

- Research papers on NLP in E-commerce
- Python NLP Libraries Documentation
- SQL Query Optimization Techniques
- Machine Learning Evaluation Metrics